

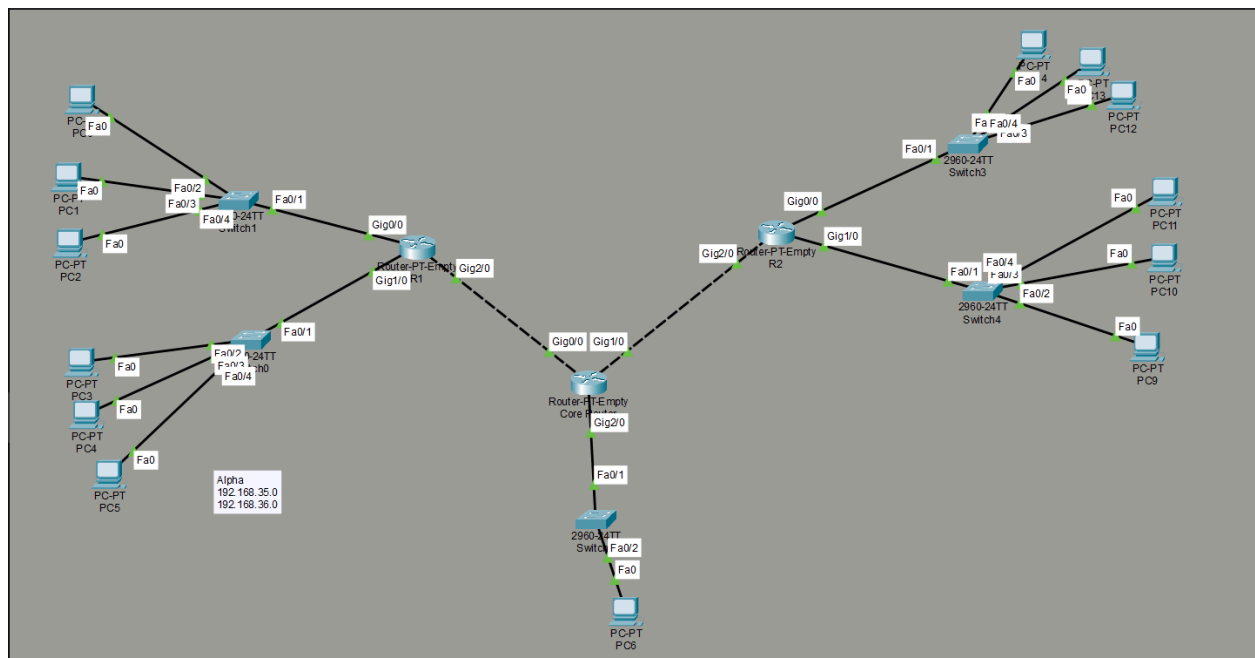
CSE 4512

Student ID: 220041135

Lab 04 Task

There are two labs for IUT CSE 2022 batch - Lab Alpha (Net Add- 192.168.id.0/24 and 192.168.id.0+1/24), Lab Beta (Net Add- 192.168.id+2.0/24, 192.168.id+3.0/24). Lab Alpha has 14 PCs connected to it and uses RIP. Lab Beta has 16 PCs connected to it and uses OSP for routing. Now, there is a core ICT router which can re-route all the routing information from these 2 labs to the ICT dept network(192.168.id+4.0/24). All the ip addresses will be assigned automatically using DHCP. Implement this scenario. id is the last two digits of your student id.

Final Topology



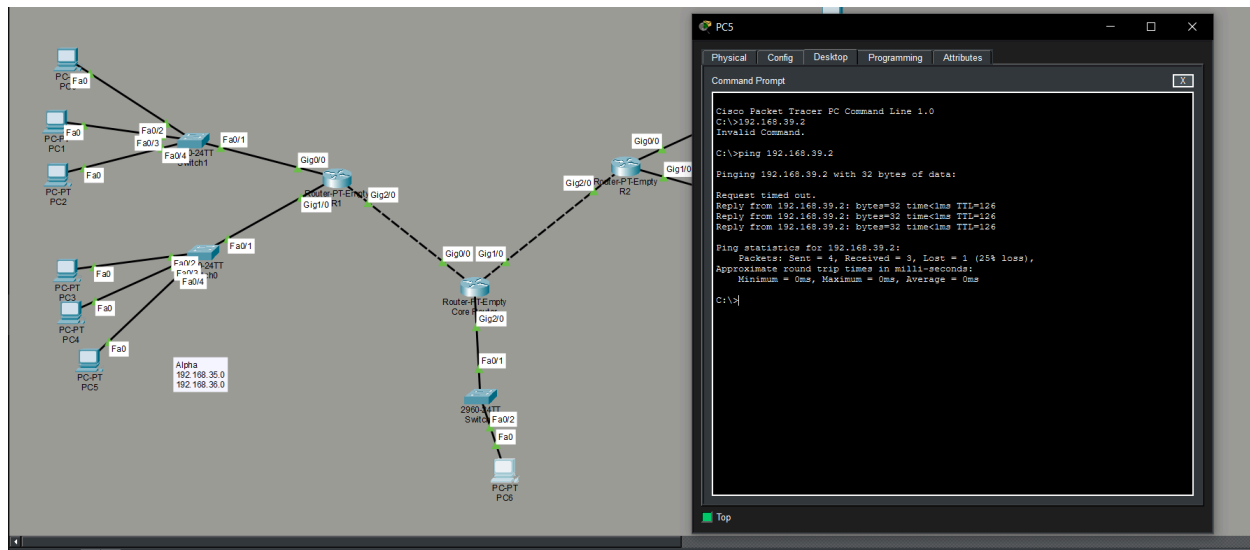
Procedure

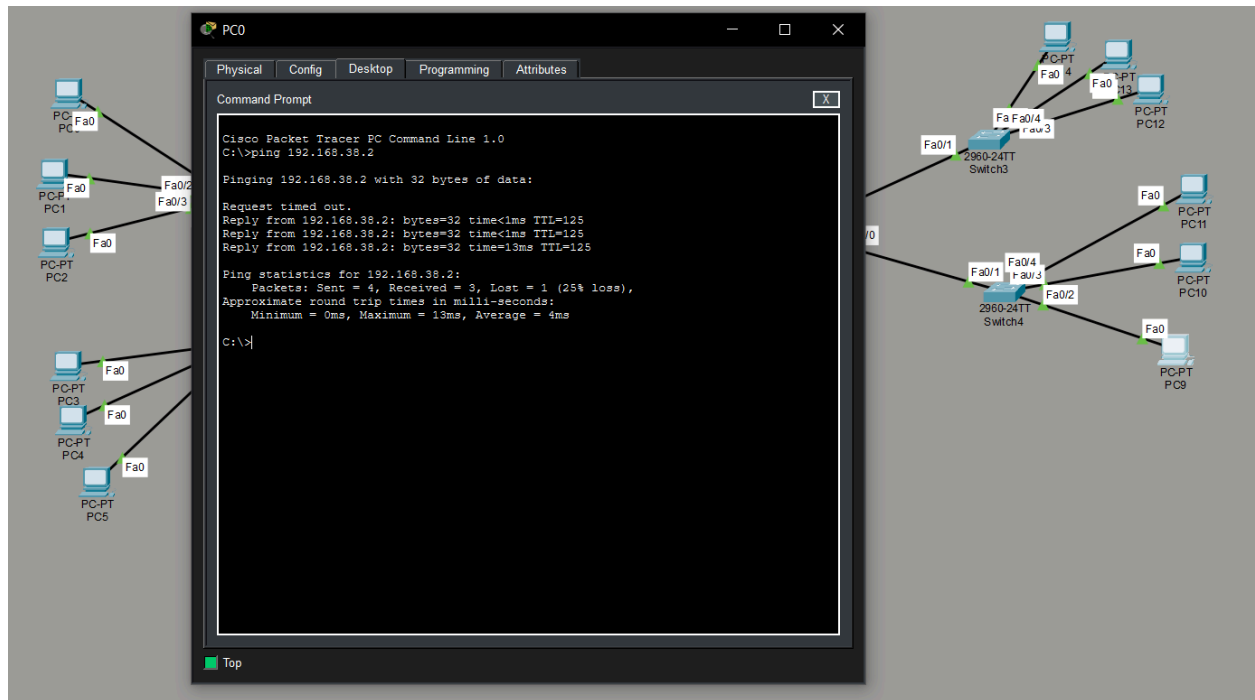
1. Firstly, I created DHCP pools in R1 and R2 for Lab Alpha and gave each connecting device DHCP addresses.

2. Then I initialized each interface based on the required network mentioned in the hypothesis.
3. For Lab Alpha, I routed R1 using RIP version 2.
4. For Lab Beta, I routed R2 using OSPF
5. For Core Router -> R1 and Core Router -> R2 connection network address of 10.10.10.0/30 and 10.10.10.4/30 was allocated respectively.
6. DHCP pool was created in Core Router for allocating DHCP address for the devices connected with the router.
7. Then the core router was also routed with rip as well ospf with R1 and R2 respectively.
8. For the communication between RIP and OSPF router was redistributed using metric 1 subnets.

Observations & Results

- Ping Checking for PCs
 - Here pings were checked for some PCs as a proof of sending packets between RIP protocol to Core Router device and from RIP to OSPF protocol.





Challenges

- Route Redistribution was confusing at first.